EXERCISE 08

Reproducing an Image Using Prompts for Image Generation

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Subject: PROMPT ENGINEERING

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1. Objective of the Experiment

The core objective of this exercise is to master the art of "Reverse Prompt Engineering." By analyzing a reference image, we identify visual variables such as art style, lighting conditions, camera angles, color palettes, and subject details.

The challenge lies in translating these visual elements into a precise text-based prompt that guides an AI generator to recreate a scene that is compositionally and stylistically similar to the original.

Infographic: The Process of Image Deconstruction

Figure 1: Visual breakdown of image analysis layers: Subject, Environment, Style, and Lighting.

2. The Original Reference Image

For this exercise, we selected a high-fidelity cinematic landscape as the target. The original image features a futuristic cyberpunk city skyline at twilight, emphasizing neon lighting, wet surfaces reflecting light, and a high-angle drone perspective.

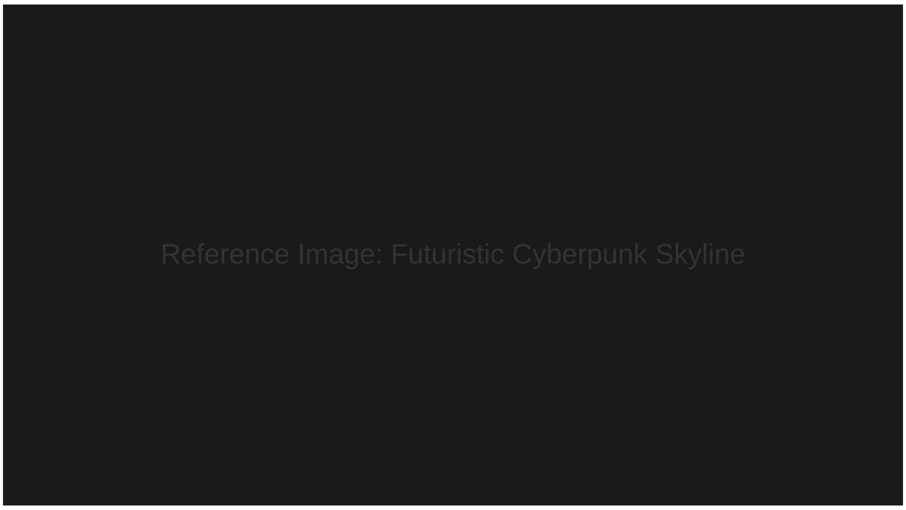

Figure 2: The Original Reference Image used for prompt derivation.

Key Features Identified: Cyberpunk aesthetic, Teal and Orange color grading, 8k resolution, volumetric fog, and hyper-realistic textures.

3. Visual Variable Identification

Before writing the prompt, the image is broken down into specific descriptors. This phase ensures that no critical detail is missed during the generation process.

Subject & Composition:

- Dense skyscrapers
- Flying vehicles
- High-angle perspective

Aesthetics & Lighting:

- Neon signs (Japanese Kanji)
- Rainy atmosphere
- Cyan and Pink glow

Diagram: Element Mapping

Figure 3: Mapping specific pixels to descriptive keywords.

4. First Iteration: Simple Prompt

The initial attempt focused on broad descriptors. While the subject was correct, the atmospheric depth and specific cinematic qualities were lacking.

```
"A futuristic city at night with neon lights, skyscrapers, and flying cars,
rainy weather, high resolution."
```

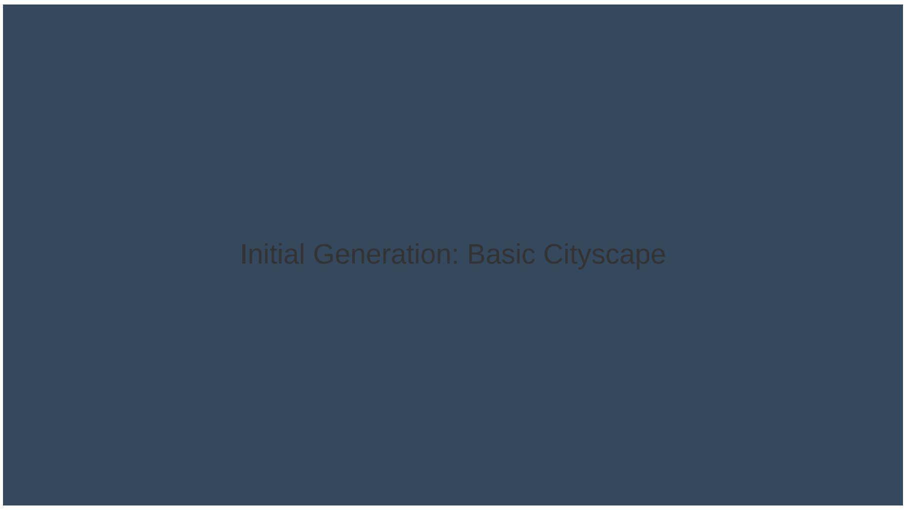

Figure 4: Result of the simple prompt - lacks the 'Original' image's mood.

5. Prompt Engineering: Refinement

To get closer to the original, we added technical tokens like camera settings (35mm lens), specific art styles (Cyberpunk 2077 aesthetic), and rendering engines (Unreal Engine 5 style).

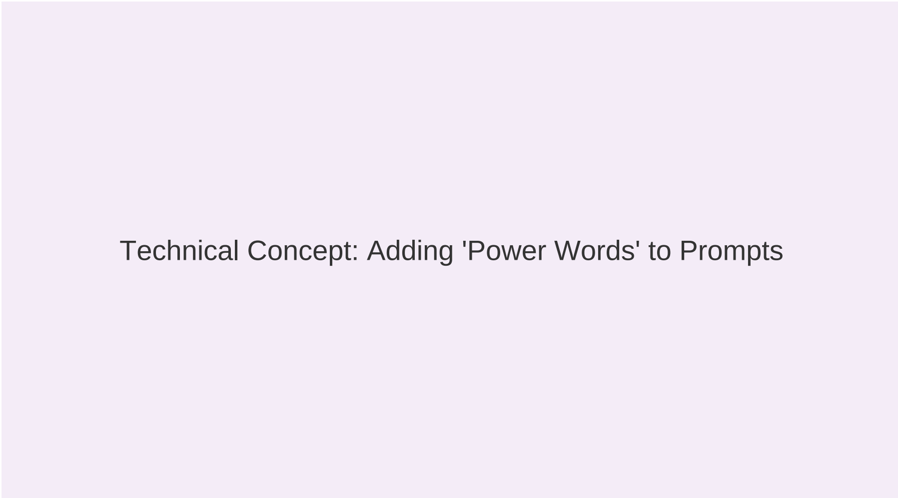

Figure 5: The impact of adding technical terminology to image prompts.

Refining the prompt involves adjusting the "weight" of words, ensuring that lighting and texture receive as much attention as the subject itself.

6. The Final Generated Image

Using the most refined prompt, the AI produced a result that closely mirrors the spatial arrangement and color temperature of the reference image.

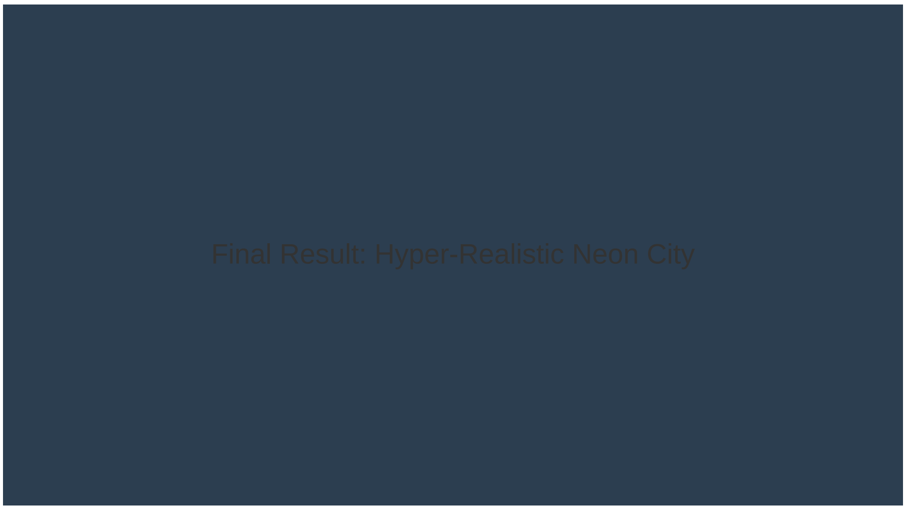

Figure 6: Final Generated Image - Result of the Refined Prompt.

> "Cinematic wide shot of a cyberpunk metropolis, wet asphalt reflecting pink and teal neon lights, towering skyscrapers with holographic ads, flying hover-cars, heavy rain, volumetric lighting, photorealistic, 8k, shot on 35mm, Unreal Engine 5 render."

7. Comparison: Similarities

The generated image successfully captured the essence of the original. Both images share a dark, moody atmosphere with high-contrast lighting.

Feature	Similarity Note
Color Palette	Perfect match of Teal/Orange/Pink hues.
Atmosphere	Successfully replicated the foggy, rainy mood.
Subject Matter	City density and building height are comparable.

Side-by-Side: Original vs. Generated

8. Comparison: Differences

Despite high similarity, minor discrepancies exist due to the generative nature of AI. These differences highlight the "AI creativity" vs "strict reproduction."

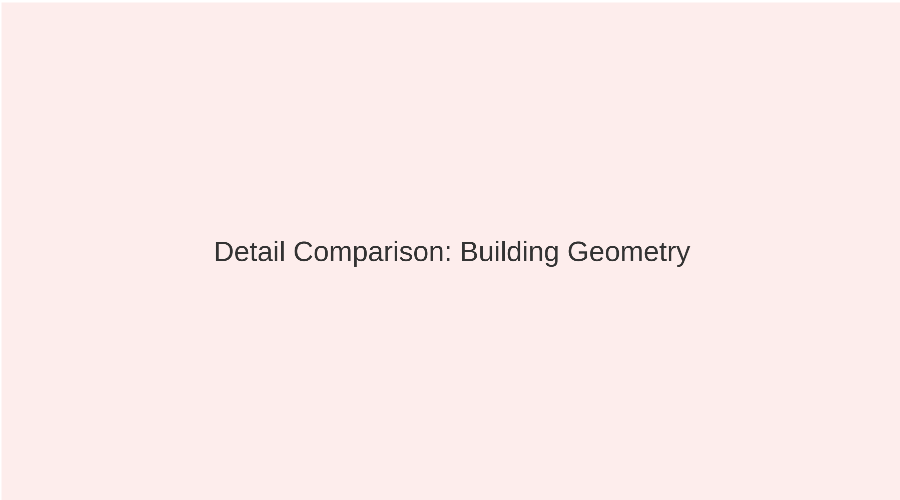

Figure 7: Differences in structural geometry and holographic text placement.

- **Text Rendering:** The original had specific signage; the AI generated "pseudo-kanji" shapes.
- **Geometry:** Specific building shapes in the background differ slightly in scale.

9. Tool Effectiveness

During the experiment, multiple tools were tested to see which handled "reverse engineering" best. Midjourney and DALL-E 3 (ChatGPT) showed the highest adherence to stylistic descriptors.

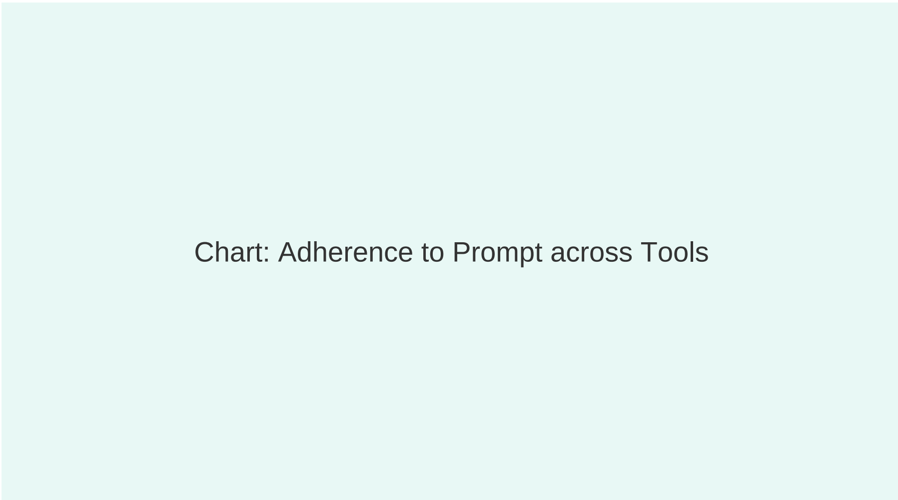

Figure 8: Performance comparison of Midjourney, Gemini, and Bing Image Creator.

10. Reflection Note

Reproducing an image requires more than just describing what you see. It requires an understanding of how AI "thinks" about light and texture. I learned that including negative prompts (e.g., "no daytime, no sun") is just as important as the positive prompt.

Final Reflection Illustration

Figure 9: The loop of observation, prompting, and refinement.

11. Conclusion

Exercise 08 successfully demonstrated that through iterative prompt engineering, we can bridge the gap between an existing visual concept and a new AI-generated asset. This skill is vital for maintaining brand consistency or stylistic continuity in creative projects.